

LAE Market Structure and Competition

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Low Altitude Economics

Week 2

Learning Objectives

- **Explain** how market structure shapes competition in different LAE services.
- **Explain** how utilization and route density can reduce average cost.
- **Identify** the main barriers that make entry costly or difficult.
- **Use** simple models to examine price competition and capacity constraints.
- **Evaluate** how Hong Kong and the wider GBA may contribute complementary capabilities.

The Battle for Kowloon

Imagine this happening right now in Hong Kong airspace:

- **Operator A** slashes prices on drone cargo flights to the New Territories, flooding the sky with last-mile deliveries.
- **Operator B** counters by securing premium air-taxi slots to the airport, promising executives they will beat the Cross-Harbour Tunnel traffic.

This is not random chaos. It is a calculated fight for survival.

- They aren't just fighting for customers. They are fighting for the rarest asset in Hong Kong: *Physical Landing Space*.
- **Quick poll (2 min):** "In 3 years, will one company own the HKIA-Central route, or will there be 10 companies competing?"

Session 1

Market Segmentation, Concentration & Economies of
Density

Concept Primer: What is "Market Structure"?

Before we look at the flying taxis, we need to understand how economists label competition.

The Three Main Structures:

- **Monopoly:** One provider serves a defined market. Its pricing power is still constrained by demand, regulation, and alternative services. (Example: A single toll tunnel).
- **Oligopoly:** A small group of big players (2 to 5 companies). They watch each other carefully. If one drops their price, the others panic. (Example: Mobile phone networks like CSL, 3, SmarTone).
- **Perfect Competition:** Hundreds of small players selling the exact same thing. Nobody has power to raise prices. (Example: Street food stalls).

LAE does not have one market structure. Competition may differ by application, corridor, and infrastructure layer.

Who Might Compete in Hong Kong's LAE Ecosystem?

- **Aircraft developers and manufacturers**
- **Transport and logistics operators**
- **Telecommunications and digital-platform providers**
- **Infrastructure providers**, including landing, charging, and maintenance facilities
- **Supporting-service providers**, including finance, leasing, insurance, certification, and professional services

*The relevant competitors depend on the specific market being analysed.
Sandbox participation alone does not establish market dominance.*

Four Market Segments — Four Different Battles

Not every low-altitude service competes the same way. The market naturally splits into four clear areas:

1. **Last-Mile Logistics:** Speed and cost rule here. Drones win when they beat road congestion on predictable B2B delivery routes.
2. **Passenger Air Taxis:** Time is the product. Premium travellers pay high prices for minutes saved in the most congested corridors.
3. **Industrial Inspection & Surveying:** Reliability matters most. Companies pay for drones that inspect bridges without shutting down ground traffic.
4. **Emergency Medical Services (EMS):** The highest-value segment. Hospitals pay whatever it takes when lives are on the line.

Concept Primer: “Market Concentration”

What is Market Concentration?

- It asks a simple question: *How much of the pie do the biggest players eat?*
- We measure this to see if a company has too much power.

Why does it matter in LAE?

- If you are the only operator on the Central-HKIA route, you can charge HK\$1,000 per ticket.
- If 5 other operators are allowed to land on your exact same rooftop, you have to drop your price to compete, or you will lose your customers.

Applied Tool: Assessing Market Power (HHI)

The Formula (Herfindahl-Hirschman Index)

$$HHI = \sum_{i=1}^N s_i^2 \quad (1)$$

where s_i is the market share (in %) of firm i in a specific corridor.

What This Actually Means for Business in Plain English

- **Why do we square the number?** Squaring makes big numbers look much bigger. It mathematically punishes markets where one player is huge. (e.g., $10^2 = 100$, but $90^2 = 8100$).
- **The Scoreboard:** HHI gives a score from near 0 (perfect competition) to 10,000 (a pure monopoly, since $100\% \times 100\% = 10,000$).
- **The Policy Use:** HHI is a screening tool. A high score may justify closer analysis, but does not by itself prove market power or determine a regulatory response.

Back-of-the-Envelope: Do we have a Monopoly?

Scenario: The government is reviewing market share on the new Northern Metropolis delivery corridor to decide if they need to step in and regulate prices.

Case A: Fragmented Market

- Four different drone operators share the airspace equally.
- Shares: 25%, 25%, 25%, 25%
- $HHI = 25^2 + 25^2 + 25^2 + 25^2$
- $HHI = 625 \times 4$
- **HHI = 2,500**
- *Result: The market is less concentrated than Case B, although four equal firms still produce an HHI of 2,500.*

Case B: Sandbox Dominance

- One early mover locked up the best landing pads before anyone else.
- Shares: 80% (Incumbent), 20% (Small Rival)
- $HHI = 80^2 + 20^2$
- $HHI = 6,400 + 400$
- **HHI = 6,800**
- *Result: The market is highly concentrated and may warrant closer examination of entry conditions and competitive conduct.*

Concept Primer: Fixed vs. Marginal Costs

To understand how flying taxis make money, you must understand two types of costs:

1. Fixed Costs (The Rent):

- Money you have to pay *even if you fly zero customers*.
- Example: Leasing a rooftop landing pad in Central costs HK\$1 Million a month, whether you fly 1 time or 1,000 times.

2. Marginal Costs (The Variable Cost):

- The tiny extra cost to fly *just one more customer*.
- Example: The electricity to charge the drone's battery for one trip across the harbour (maybe HK\$50).

From Concept to Math

Fixed and marginal costs are the foundation. The next step is to combine them into one powerful number that operators use every day:

Average Cost per Flight

This tells us the true cost of each flight when we spread the huge fixed costs across many trips.

Applied Tool: Average Cost per Flight

The Formula

$$ATC = \frac{\text{Fixed Costs}}{\text{Number of Flights}} + \text{Marginal Cost} \quad (2)$$

What This Actually Means for Business

- Fixed Costs (e.g. rooftop rent) stay the same no matter how many times you fly.
- The more flights you operate, the lower the Fixed Cost per flight becomes.
- Marginal Cost is the extra cost of one additional flight (mainly electricity and minor wear).

Back-of-the-Envelope: How Fixed Costs Make Low Utilization Expensive

Scenario: A rooftop landing pad in Central costs HK\$1,000,000 per month to lease. Marginal cost (electricity + minor maintenance) = HK\$50 per flight.

Low Utilization (20 flights/day)

- Fixed cost per day HK\$33,333
- Fixed cost per flight: $\$33,333 / 20 = \text{HK\$1,667}$
- Marginal cost: HK\$50
- **Total cost per flight: HK\$1,717**
- *Ticket price must be very high — only premium customers can afford it.*

High Utilization (200 flights/day)

- Fixed cost per day HK\$33,333
- Fixed cost per flight: $\$33,333 / 200 = \text{HK\$167}$
- Marginal cost: HK\$50
- **Total cost per flight: HK\$217**
- *Ticket price becomes competitive and profitable.*

Takeaway: High fixed costs make low utilization deadly. Operators must maximise flights per day to survive.

Concept Primer: The "Density" Secret

Why do traditional airlines want bigger planes?

- Airlines rely on **Economies of Scale**. It's cheaper per person to fly one giant Boeing 777 with 300 seats than to fly 15 small planes with 20 seats.

The LAE Shift: We can't build bigger drones.

- Flying taxis are stuck at 4 seats. Delivery drones carry 5kg. We can't scale the vehicle.
- Instead, we rely on **Economies of Density**.
- Since your Fixed Cost (the rooftop rent) is massive, flying 10 times a day is a financial disaster. Higher route volume can spread fixed costs across more flights, although demand, capacity, congestion, weather, and regulation limit achievable density.

Applied Tool: The Density Cost Advantage

The Formula

$$\text{Average Cost per flight (AC)} = \frac{\text{Fixed Route Costs}}{\text{Flights} \times \text{Density}} + \text{Marginal Cost} \quad (3)$$

What This Actually Means for Business

- **Why Density Wins:** If density (the number of flights) is the denominator, then running maximum flights through a single corridor drastically lowers the Average Cost of every single ticket.
- **Potential Incumbent Advantage:** An established operator with greater route volume may sustain a lower average cost than a small entrant.
- **Network Effects:** A broader network may become more useful to customers, although this is conceptually different from economies of density.

Back-of-the-Envelope: The Density Trap

Scenario: Calculating the lowest price we can charge to break even.

Assumptions: A rooftop landing pad (Fixed Cost) is HK\$50,000/day to rent. Electricity/wear (Marginal Cost) is HK\$100/flight.

Operator A: Low Density

- Volume: Only 20 flights per day.
- Pad rent spread out: $\$50,000 / 20 = \$2,500$
- Electricity: \$100
- **Total Cost to Operator: HK\$2,600/flight**
- *Result: Operator A must charge HK\$3,000 a ticket just to survive. Only billionaires will fly.*

Operator B: High Density

- Volume: 500 flights per day.
- Pad rent spread out: $\$50,000 / 500 = \100
- Electricity: \$100
- **Total Cost to Operator: HK\$200/flight**
- *Result: Operator B has a lower average cost under these assumptions, although feasibility and market price still depend on demand and capacity.*

Session 2

The Moat: Entry Barriers & Defense

Entry Barriers: The NASA Framework

- In 2018, NASA studied why new drone and flying-taxi companies struggle to enter the market.
- They created a simple, practical list of the biggest obstacles.
- We still use this list today because it is clear and directly applies to Hong Kong.
- These barriers create a huge “moat” around the LAE market.
- They make it very difficult and expensive for new startups to start operating.

The Three Main NASA-Style Entry Barriers

1. **Regulatory / Certification (The Valley of Death)** Companies must spend millions on lawyers and safety tests for years before they are legally allowed to fly paying customers.
2. **Infrastructure Availability** Finding physical rooftop space and heavy electrical connections in land-scarce Hong Kong is extremely difficult for a new company.
3. **Public Perception** A single neighbourhood noise complaint or privacy concern can force the government to shut down a corridor overnight.

The importance of each barrier varies by application and may change as policy, technology, and the market develop.

Concept Primer: "Sunk Costs"

What is a Sunk Cost?

- It is money you have already spent that you **can never get back**, even if you quit the business.
- If you buy a van for a delivery business and then quit, you can sell the van to get some money back. That is NOT a sunk cost.
- But if you spend HK\$10 Million paying lawyers to get a flight permit, and then you quit... you cannot sell that permit to anyone else. The unrecoverable portion of that expenditure is a sunk cost.

Why it matters:

- High Sunk Costs terrify new startups. If a startup knows they have to burn HK\$10M just to enter the market, they will probably stay away. Big companies use this fear to their advantage.

Concept Primer: "Limit Pricing"

How do you defend your turf if you are the first one in?

- Let's say you are the only flying taxi in Hong Kong. You *could* charge HK\$5,000 a ticket and make huge profits today.
- But if your profits are that huge, rich investors in another country will see that, buy their own drones, pay the Sunk Costs, and invade your market.

The Strategy: Limit Pricing

- You intentionally do not charge the absolute maximum.
- You charge a price that makes you a *little* bit of money, but is perfectly calculated to guarantee that any new competitor would lose money if they tried to copy you.

Applied Tool: Limit Pricing (Defending a Route)

The Formula

$$P_{\text{Limit}} = AC_{\text{Entrant}} = \frac{\text{Sunk Entry Cost}}{\text{Expected Volume (Q)}} + MC \quad (4)$$

What This Actually Means for Business

- **Weaponising the NASA Barriers:** The NASA barriers (Regulation, Infrastructure) create massive "Sunk Entry Costs" for new rivals. They have to pay millions just to show up to the starting line.
- **The Incumbent's Game:** If you own the route, you charge a price (P_{Limit}) that is exactly equal to the Average Cost (AC) of a new entrant.
- **The Core Lesson:** This simplified calculation estimates an entrant's break-even price. It does not guarantee that limit pricing will prevent entry.

Back-of-the-Envelope: Defending the Corridor

Scenario: You run a monopoly eVTOL route. A rival wants to enter. They expect to capture 10,000 flights. Their Marginal Cost (MC) is \$200. How do you price your tickets to keep them out?

Case A: High NASA Barriers

- CAD forces rival to spend years building their own vertiport.
- Sunk Entry Cost: HK\$10 Million
- $AC_{\text{Entrant}} = \frac{10,000,000}{10,000} + 200$
- $AC_{\text{Entrant}} = 1000 + 200 = 1200$
- **Your Limit Price: HK\$1,190**
- *Result: You can safely charge high prices. The rival cannot match \$1,190 without losing money.*

Case B: Low Barriers (Shared Hubs)

- CAD forces you to share your landing pad with the rival.
- Sunk Entry Cost drops to: HK\$1 Million
- $AC_{\text{Entrant}} = \frac{1,000,000}{10,000} + 200$
- $AC_{\text{Entrant}} = 100 + 200 = 300$
- **Your Limit Price: HK\$290**
- *Result: Your moat is gone. You must slash your prices to \$290 to stop the rival from invading.*

Illustrative assumptions for classroom analysis; not current market data.

Hong Kong Experimentation and GBA Scale

- **GBA capabilities:** Strong aircraft, drone, electronics, battery, and supply-chain capabilities.
- **Hong Kong's possible contribution:** A complex urban environment together with finance, insurance, logistics, legal, and professional services.
- **Sandbox X:** Controlled trials can generate evidence, reduce uncertainty, and refine project concepts and regulatory requirements.
- **Strategic opportunity:** Combine complementary capabilities while addressing cross-boundary procedures, customs, data, communications, and safety.

Hong Kong's advantage is a strategy to build, not an automatic outcome.

Session 3

Price Wars & Co-opetition

Concept Primer: The "Race to the Bottom"

What happens when two airlines fly the exact same route?

- If Cathay Pacific and HK Express both have half-empty planes going to Tokyo, what do they do? They drop the price to try and steal the other's customers.
- They will keep undercutting each other until the ticket price is exactly equal to the cost of the fuel and the sandwich. All profit disappears.
- Economists call this a **Bertrand Price War**.

Applied Tool: Bertrand Price Competition

The Formula

$$P = MC \quad (\text{if capacity is unconstrained}) \quad (5)$$

What This Actually Means for Business

- **The Death of Margins:** If operators have unlimited landing space and empty seats, they will undercut each other until Price (P) equals Marginal Cost (MC). No one makes a profit.
- **Capacity Saves the Day:** Because HK real estate restricts vertiports to 1 or 2 pads, neither operator can serve the whole market.
- **The Core Lesson:** Capacity constraints may soften price competition, while reliability, timing, access, and service quality can also differentiate operators.

Back-of-the-Envelope: Capacity Saves the Margin

Scenario: Operator A and Operator B fly the same route. Electricity/Cost per flight is \$100. Customer Willingness to Pay is \$500. Total Demand is 100 passengers.

Unconstrained Capacity

- Both operators have huge fleets.
- Operator A charges \$400.
- Operator B drops price to \$300 to steal all 100 passengers.
- Operator A counters at \$200.
- **Final Price drops to \$100 (The cost to fly).**
- *Result: A price war destroys the industry. Zero profit.*

Geofenced / Capacity Constrained

- Vertiport rules limit both operators to only 30 flights a day each.
- Total supply (60) is less than total demand (100).
- Operator A charges the max: \$500.
- Operator B doesn't need to undercut; they can easily fill their 30 seats at \$500 anyway.
- **Price pressure may be weaker, but the final price need not equal maximum willingness to pay.**
- *Result: Limited capacity may support margins, but demand, allocation rules, regulation, and differentiation still matter.*

The Rise of "Co-opetition" (Frenemies)

Aviation Benchmark: Airlines compete for ticket sales, but they share Oneworld lounges and codeshare flights to survive.

LAE Shift: Operators must share fundamental infrastructure to survive the early market phase.

- **What is Co-opetition?** Competing fiercely on some fronts while cooperating on others to save money and grow the overall market.
- **Shared Vertiports:** Sharing a roof reduces your fixed costs and builds the network faster.
- **Strategic Choice:** You compete on the *vehicle app experience* and brand, but you cooperate on the *landing pads and digital traffic control*.

Applied Tool: Co-opetition Cost Saving Formula

The Formula

$$\text{Savings per Operator} = \frac{\text{Fixed Infrastructure Cost} \times (N - 1)}{N} \quad (6)$$

What This Actually Means for Business

- N = number of operators sharing the same vertiport or UTM system.
- Fixed cost (e.g. rooftop rent) is split among more companies.

Back-of-the-Envelope: Co-opetition Savings

Scenario: A prime Kowloon rooftop vertiport costs HK\$100,000 per day to lease and maintain.

Alone (No Sharing)

- 1 operator pays full rent.
- Fixed cost per day: **HK\$100,000**
- If flying 100 times/day: HK\$1,000 per flight just for rent.

Sharing with 4 Operators

- 4 operators split the same rooftop.
- Fixed cost per operator: $\$100,000 / 4 = \mathbf{HK\$25,000}$
- Savings per operator = HK\$75,000 per day.
- If each flies 100 times/day: rent cost drops to HK\$250 per flight.

Takeaway: Sharing infrastructure is one of the fastest ways for early LAE operators to survive Hong Kong's extreme land costs.

Illustrative assumptions for classroom analysis; not current market data.

Global Regulatory Comparators

How does Hong Kong's strategy compare to the rest of the world?

- **Hong Kong (Sandbox X):** Controlled trials test project concepts, operating conditions, infrastructure, and cross-boundary procedures.
- **Europe (EU U-space):** Highly structured, digital-first. They are building the software rules before they allow the hardware to fly.
- **USA (FAA Innovate28):** An implementation plan coordinating certification, operations, airspace, infrastructure, environment, and community engagement.
- **Managerial lesson:** Firms must adapt their entry strategies to how each jurisdiction sequences trials, certification, infrastructure, and deployment.

In-Class Exercise: Market Simulation

Activity: Oligopoly Pricing Game

- **Format:** Groups of 4–5
- **Scenario:** You are one of two operators granted access to a new Northern Metropolis drone delivery corridor.
- **Task:** Using the provided handout:
 1. Look at the Payoff Matrix.
 2. Decide if your team will set a "High Price" or a "Low Price".
 3. What happens to your profits if the other team cheats?
- **Timing:** 15 minutes group work + 5 minute share-out.

Exercise Template (Visual)

[Distribute 1-page printable simulation template]

The Payoff Matrix (Team A vs Team B):

If both price High → Both survive with healthy margins.

If one prices Low → Captures market, but margins evaporate.

Discussion: How does a physical capacity constraint (only 1 landing pad) change the math of this game?

Key Takeaways

- LAE market power is influenced by **infrastructure scarcity** (vertiports) and **route density**.
- NASA barriers to entry are immense; big operators use these massive Sunk Costs to establish **Limit Pricing**, keeping rivals out.
- Capacity constraints and service differentiation may soften price competition, but they do not guarantee high prices or profits.
- **Co-opetition** may reduce duplicated infrastructure cost, but requires careful rules for access, investment, and control.

Week 2 Assignment Briefing

Two-page Market Competition Memo (max 800 words)

- **Due:** Beginning of Week 3
- **Task:** Act as an operator entering a specific HK corridor.
- **Requirements:**
 1. Give a rough score for market concentration (HHI).
 2. Explain how you will achieve "Density" on this specific route.
 3. Classify the top 3 entry barriers (NASA-style) keeping your rivals out.
 4. Recommend one strategic "co-opetition" move to save money.
- **Grading Focus:** Applying the economic concepts (40%), realistic HK linkage (30%), simple math quantification (20%), professionalism (10%).

Cost Structures and Revenue Management

Preview question for next week:

How do you stop your business from going bankrupt when a massive Hong Kong rainstorm grounds your entire fleet for 48 hours?

Thank You + Q&A

Pre-Class Readings Reminder (for Week 3):

- *McKinsey: Unit Economics of eVTOLs*
- *GBA Aviation Infrastructure Cost Case Study*

Questions?

(Final 15 minutes reserved for extended Q&A)